

Theme: Cyber-threats

I. Context Setting

In 2026, the digital battlefield has shifted from human-led breaches to "**Agentic Cyber-Warfare**." With the emergence of Ransomware 5.0—fully autonomous AI chains that probe, exploit, and negotiate ransoms without human intervention—traditional perimeter defenses have become obsolete. This theme is of paramount importance because the "dwell time" (the window between infection and detection) has shrunk from days to mere minutes. Today's industry relevance lies in **Defensive Autonomy**: using AI agents to fight AI attackers. As organizations face "Shadow Agent" risks—where trusted internal AI agents are manipulated via memory poisoning or prompt injection—the challenge is no longer just blocking IPs, but verifying the *intent* of every autonomous action within the network.

II. EXAMPLE Applications

- **Autonomous Honeypot Deployment:** Use AI to dynamically generate and spin up "decoy" cloud environments that mimic real high-value databases the moment a scanning agent is detected on the network.
 - **Sleeper Agent Discovery:** Build a specialized "Auditor Agent" that scrapes internal system logs and execution traces to identify "memory poisoning" where an attacker has surreptitiously altered an AI agent's long-term instructions.
 - **Predictive Threat Feed Synthesis:** Scrape specialized dark-web forums and Telegram channels using NLP to identify "Zero-Day-as-a-Service" trends, automatically updating firewall rules before the exploit hits the mainstream.
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III. EXAMPLE Workflow

The team builds a **Python-based ingestion engine** that scrapes the "CVE Details" database and several cybersecurity subreddits for emerging exploit Proof-of-Concepts (PoCs). These signals are processed by an event-driven **Google Cloud Function** that uses **Mistral Devstral** to assess if the organization's current tech stack is vulnerable. High-risk findings are stored in **BigQuery**, which serves as the "Threat Brain," while a **Django-based "War Room"** dashboard visualizes the live attack surface. If a critical match is found, the system triggers an automated **Cloud Run** script to patch the

vulnerable container or alert the SOC (Security Operations Center) via a prioritized Slack payload.

IV. End-to-End Hints

- **Focus on "Mean Time to Respond" (MTTR):** In a cyber-threat hackathon, speed is the primary KPI; your project should prove how AI can reduce the time between "Threat Identified" and "Threat Mitigated" by at least 50% compared to a human analyst.
- **Verify the "Chain of Reasoning":** Don't let your security agent be a "black box." Ensure your dashboard displays the agent's *thought process* (why it flagged an activity as malicious), as this audit trail is essential for CISO approval and regulatory compliance.